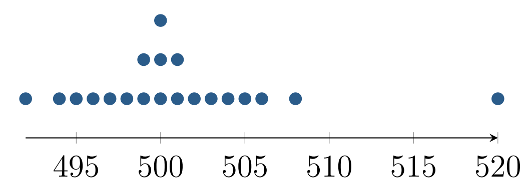
One Sample Far From the Others

Unit 2 · Topic 2.12 · TODAY'S PROBLEM

Suppose a class uses a simulator containing a very large population of package weights with population mean $\mu = 500$ grams. Each of 20 students independently takes a random sample of $n = 10$ packages and records the sample mean $\bar{x}$. The dotplot shows the 20 sample means. Elena's sample mean is 520 grams. She says, "My sample must be wrong because its mean is far from everyone else's." Another student says, "A random sample can be unusual without being collected incorrectly."



Sample mean package weight (grams), n=10

FIRST TAKE (5 min, on your sheet)

Does the dot at 520 prove Elena made a mistake? Give one reason from the picture.

DOK 1 **(a)** Identify the population parameter and Elena's sample statistic, including symbols, values, and units.

DOK 2 **(b)** Describe the center and overall range of the sample means, and the gap next to 520 grams. Two sentences are enough.

DOK 3 **(c)** In 3–4 sentences, decide whether the dotplot proves Elena's sample was collected incorrectly. Use the spread or gap as evidence. Explain how the population mean can stay fixed while correctly collected sample means differ.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.